Interpretable third-order penalized least-squares trend estimation: known-jerk simulations and real vehicle trajectories

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ARTICLE INFO

Keywords:
Penalized least squares
third differences
vehicle jerk
smoothness index
truncated singular-value decomposition
smoothing splines
NGSIM

ABSTRACT

The third difference of recorded vehicle position approximates jerk but strongly amplifies observation noise. We investigate the established third-order penalized least-squares (PLS) trend model, its reference third-difference mean and the normalized, interpretable smoothness index S, with a focus on *physical reconstruction and forecasting*, not solver execution time. In the independent Gaussian variance-ratio model, S equals the average spectral fraction of the observed third-difference covariance attributable to measurement noise. A controlled relative-noise calibration experiment with 1,944 paired noise–jerk conditions (72 independent innovation parents) finds Spearman association 0.965 and mean absolute calibration error 0.0115 for joint-REML-selected S under correct covariance assumptions, while correlated observation noise substantially worsens calibration. We also compare PLS with hard truncated-SVD filtering of the same third-difference operator, GCV cubic smoothing splines and Savitzky–Golay filtering using 1,296 archived known-jerk synthetic trajectories. PLS with a historical cubic-OLS reference and Gaussian profile-likelihood selection attains historical discrete-jerk RMSE 2.441 m/s^3, versus 2.942 m/s^3 for SVD truncation at nominally matched fixed-reference complexity. Real-data forecasting is assessed separately on 200 quality-qualified Federal Highway Administration NGSIM episodes, where the PLS and matched-SVD three-point continuations are comparable and a GCV smoothing spline with quadratic Taylor continuation achieves the smallest pooled recorded-position window RMSE. Thus third-order PLS provides a physically interpretable, tunable estimator for noisy jerk, but S is a *relative, model-dependent* noise index rather than absolute sensor precision, and no estimator dominates every physical or forecasting target.

1. Introduction

Vehicle trajectories provide a physical interpretation for higher-order trend smoothing: the first three time derivatives of longitudinal position are velocity, acceleration and jerk. The US Federal Highway Administration's Next Generation Simulation (NGSIM) dataset reports camera-derived vehicle positions every 0.1 seconds (U.S. Department of Transportation, Federal Highway Administration, 2016). Earlier vehicle-trajectory research emphasizes that estimating acceleration from these observations magnifies measurement noise (Thiemann, Treiber and Kesting, 2008); third derivatives amplify it further. This motivates examining a third-difference penalty ($d = 3$), rather than restricting the study to the commonly used second-difference penalty ($d = 2$).

We study the PLS model of Guerrero (2007), its trace-based smoothness index (Guerrero, 2008) and classical tuning criteria (Cortés-Toto, Guerrero and Reyes, 2017). The present manuscript distinguishes three questions: whether normalized smoothness quantifies relative spectral observation noise under a stated covariance model, whether a third-order fit recovers known physical jerk, and whether a separate continuation rule predicts future camera-recorded positions. Companion papers examine numerical accuracy and computational complexity. We therefore combine rigorously controlled analytic-jerk/noise simulations with chronological comparisons on archived observed vehicle trajectories.

We also compare two boundary continuation rules for the same fitted $d = 3$ PLS trend and benchmark them against natural cubic smoothing splines with GCV-selected penalties, Savitzky–Golay differentiation and local polynomial fits. These are standard local/global families for estimating derivatives from noisy experimental data (Ahnert and Abel, 2007), rather than independent numerical matrix-factorization methods. The resulting analysis tests an implementation and application of an established PLS model, not an assertion of historically first use of $d = 3$ differences.

This application sits within a substantial prior literature. Whittaker–Henderson penalization predates the present study (Whittaker, 1923), and extrapolation from penalized smoothers has been developed in modern statistical treatments (Biessy, 2026). For measured vehicle motion, Marczak and Buisson (2012) examined NGSIM trajectory filters and jerk-related diagnostics, while Cao, Bryant, Molteno, Fox and Parry (2021) proposed trajectory splines using position and velocity data. Physically constrained quadratic-programming methods (Fan, Jin and Yin, 2024) and continuous-time jerk-driven stochastic motion priors (Tang, Yoon and Barfoot, 2019) address related but distinct problems. Our unconstrained, position-only third-difference PLS is neither a substitute for kinematic feasibility constraints nor a new general forecasting model. The contribution here is a transparent comparison of its reference-jerk and smoothing choices, conditional calibration, and reconstruction versus prediction targets.

The real NGSIM dataset supplies camera-derived position, but not independently measured latent truth or jerk. We do not contaminate the real positions to create artificial validation labels. Separately, in a fully documented synthetic factorial experiment, we prescribe clean positions and known jerk before adding controlled measurement errors;

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only those simulated trajectories support direct calibration and true-jerk reconstruction assessments.

1.1. The noise amplification motivating third-order smoothing

If measurement noise ε_t has independent variance σ_ε^2, third forward differentiation uses stencil $(-1, 3, -3, 1)$, hence

$$\operatorname{Var}\left(\frac{D_3\varepsilon_t}{(\Delta t)^3}\right) = \frac{20\sigma_\varepsilon^2}{(\Delta t)^6}. \tag{1}$$

With $\Delta t = 0.1$ s, direct third differences can be extremely unstable. Equation (1) is a conditional mathematical illustration, not an empirical noise-variance estimate: camera-tracking errors may be temporally correlated. Regularization suppresses variation but can bias sudden real changes in jerk. Thus plotted derivatives remain *estimates*, not ground truth.

2. Third-order PLS trend estimation and physical comparisons

2.1. Observation model and third-order difference penalty

Let $y = (y_1, \dots, y_n)^\top$ be position measurements in meters recorded at interval $\Delta t = 0.1$ s and let τ be their unknown smooth historical trajectory. The $(n-3) \times n$ difference matrix D_3 uses the stencil $(-1, 3, -3, 1)$; $D_3\tau/(\Delta t)^3$ is a discrete jerk proxy in m/s^3. Following Guerrero (2007), we estimate the trajectory by minimizing

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y - \tau\|_2^2 + \lambda \|D_3\tau - \mu\mathbf{1}\|_2^2 \right\}, \tag{2}$$

where μ is the reference *discrete* third difference, and $\lambda \geq 0$ controls regularization. Its physically scaled reference jerk equals $\mu/(\Delta t)^3$. Writing $Q = D_3^\top D_3$, the unique finite-λ solution is

$$\widehat{\tau}_{\lambda,\mu} = (I + \lambda Q)^{-1}(y + \lambda\mu D_3^\top \mathbf{1}). \tag{3}$$

The $d = 3$ polynomial nullspace has dimension three: unpenalized quadratic position, linear velocity and constant acceleration. At strong regularization, $\mu = 0$ imposes a quadratic limit, whereas fixed $\mu \neq 0$ imposes a cubic position trajectory with constant discrete third difference. These *limits* are not assumptions that real vehicles have constant jerk throughout the observed interval.

2.2. An interpretable, normalized smoothness index

Let $\delta_1, \dots, \delta_m > 0$ be the positive eigenvalues of Q, with $m = n - 3$. The smoothness index used throughout the `trendestimation` library is

$$S(\lambda) = \frac{1}{m}\sum_{j=1}^{m} \frac{\lambda\delta_j}{1 + \lambda\delta_j} \in [0, 1]. \tag{4}$$

It is monotone in λ and therefore provides an interpretable, comparable normalized regularization coordinate. For a fixed reference μ and a fixed penalty λ, let $H_\lambda = (I + \lambda Q)^{-1}$; then

$$\mathrm{df}(\lambda) = \operatorname{tr}(H_\lambda) = n - mS(\lambda). \tag{5}$$

This *conditional* degrees-of-freedom formula must not be applied unchanged when μ is estimated from the same observations. Under the independent Gaussian variance-ratio model, the smoothness index also admits the relative-noise interpretation tested in Section 4; it is not the absolute measurement precision of a sensor.

2.3. Smoothing, reference jerk and choice of regularization

We compare the existing PLS policies: a fixed reference $\mu = 0$; the sample mean of the noisy third differences; a global cubic polynomial fitted to *historical* positions (its discrete third difference gives μ); and a jointly estimated reference from Gaussian restricted likelihood. For the stochastic interpretation, $\lambda = \sigma_\varepsilon^2/\sigma_u^2$ is the ratio of observation noise variance to genuine discrete third-difference innovation variance. The Gaussian interpretation adds distributional assumptions beyond the second-moment PLS construction of Guerrero (2007). When μ is estimated rather than fixed, the mean-adjusted restricted likelihood must also eliminate its nuisance mean direction.

We retain the library's GCV and Gaussian profile-ML choices and explicitly separate *fixed-μ profile likelihood* from *joint unknown-μ REML*; they need not minimize the same physical loss. The original historical case also screened LOOCV, AICc and restricted residual BIC, as shown in its archived selection comparison. Here they are background sensitivity analyses, not a paper about computational methods or root-finding performance. All tuning uses the training portion, never the held-out future.

2.4. Truncated SVD as an *estimator* baseline

A full singular-value decomposition used solely to evaluate Eq. (3) is a numerical implementation of *the same PLS estimator*, not a competitor. For a meaningful comparison we instead use *hard spectral truncation* of the *same* third-difference operator. Write

$$D_3 = U\ \mathrm{diag}(\sigma_1, \dots, \sigma_m)V_+^\top, \qquad 0 < \sigma_1 \leq \cdots \leq \sigma_m, \tag{6}$$

where the V_+ columns are ordered from lowest to highest difference roughness, and let V_0 span $\ker D_3$. Choose q satisfying $D_3 q = \mu\mathbf{1}$, for example $q_i = \mu\binom{i-1}{3}$. PLS may be written as a *soft spectral filter*:

$$\widehat{\tau}_{\mathrm{PLS}} = q + V_0V_0^\top(y - q) + \sum_{j=1}^{m} \frac{v_j v_j^\top (y - q)}{1 + \lambda\sigma_j^2}. \tag{7}$$

The truncated-SVD comparison is instead the *hard cutoff*

$$\widehat{\tau}_{\mathrm{TSVD},r} = q + V_0 V_0^{\top}(y-q) + \sum_{j=1}^{r} v_j v_j^{\top}(y-q), \quad 0 \le r \le m. \tag{8}$$

This is a genuine, distinct low-roughness spectral smoother. We specify two training-only comparisons: (i) choose r by conditional GCV over ranks $0, \dots, m-1$; (ii) nominally match the conditional PLS complexity with $r = \mathrm{round}(\mathrm{df}_{\mathrm{PLS}} - 3)$. The latter is a matched-complexity *ablation*, not an independently optimized SVD procedure. Both use the same training-only global cubic reference μ as the corresponding profile-ML PLS comparator. Their normalized hard-filter index is $S_{\mathrm{TSVD}} = 1 - r/m$.

Conditioning on the estimated reference. With P_r the projector onto the $3 + r$ retained modes, the implemented TSVD rank criterion is

$$\mathrm{GCV}_{\mathrm{conditional}}(r) = \frac{n\|(I - P_r)(y - q_{\widehat{\mu}})\|^2}{[n - (3+r)]^2}, \qquad 0 \le r < m. \tag{9}$$

It uses the training-estimated cubic $\widehat{\mu}$ in the residual but treats that reference as fixed in the denominator $n-(3+r) = m - r$. It is therefore *conditional plug-in GCV*, not exact GCV for the entire data-dependent fitting map. In particular, retaining r near m leaves few nominal residual degrees of freedom.

The same issue makes effective-complexity matching *nominal* rather than exact. Write $\widehat{\mu} = a^{\top} y$, where a is the linear cubic-OLS reference functional, $q_{\mu} = \mu q_1$ with $D_3 q_1 = \mathbf{1}$, and let P_r project onto the $3 + r$ retained TSVD directions. For fixed λ and r (ignoring their additional data-driven selection), the linear fits have effective degrees of freedom

$$\mathrm{df}_{\mathrm{PLS}}^{\mathrm{plug}} = \mathrm{tr}(H_{\lambda}) + a^{\top}(I - H_{\lambda})q_1, \tag{10}$$

$$\mathrm{df}_{\mathrm{TSVD}}^{\mathrm{plug}} = 3 + r + a^{\top}(I - P_r)q_1. \tag{11}$$

A fixed-rank GCV denominator for the complete plug-in linear map would instead be $n - \mathrm{df}_{\mathrm{TSVD}}^{\mathrm{plug}}$. The rank matching in Experiment008 equates only the fixed-reference trace components; it does not equate these full plug-in degrees of freedom. Both reference estimates are based on historical data only. Classical truncated singular-value regularization has a broader inverse-problem literature; we do not claim the hard-cutoff construction is new (Renaut, Helmstetter and Vatankhah, 2019).

2.5. Other physical jerk and trajectory baselines

The baseline families are (i) natural cubic smoothing splines with their second-derivative roughness penalty chosen by training-only GCV, (ii) Savitzky–Golay local polynomial smoothing (window 21, degree three), (iii) ordinary local polynomial and constant-velocity or constant-acceleration extrapolations, and (iv) raw third differences as an intentionally noisy derivative reference (Savitzky and Golay, 1964; Ahnert and Abel, 2007; Cao et al., 2021). A classical cubic spline penalizes the *second* derivative, whereas Eq. (2) penalizes *deviations of the third difference from* μ; these are different physical smoothness assumptions. The comparison measures historical discrete jerk only when a synthetic trajectory supplies known truth, and future position when a chronological future is available.

2.6. Forecast continuation is separate from historical smoothing

Given a fitted historical trend, the primary $d = 3$ continuation assumes constant future third difference equal to the chosen reference μ. At future integer horizon h,

$$\widehat{\tau}_N(h) = \widehat{\tau}_N + h\nabla\widehat{\tau}_N + \binom{h+1}{2}\nabla^2\widehat{\tau}_N + \binom{h+2}{3}\mu. \tag{12}$$

This is the algebraic $d = 3$ extension of the lower-order forecasting construction of Guerrero (2007), *not a new probabilistic prediction model*. A secondary continuation fits a quadratic to the last 25 fitted positions after subtracting the polynomial particular solution q, then adds the future q. Both schemes use only historical data; neither can anticipate future changes in driver behavior. A separate spline baseline uses a quadratic Taylor continuation from spline endpoint derivatives. The pooled forecast error at horizon $H \in \{5, 10, 20\}$ is a *cumulative evaluation-window RMSE*, not an error at only the final future frame:

$$\mathrm{RMSE}_H = \left\{ \frac{1}{NH} \sum_{i=1}^{N} \sum_{h=1}^{H} \left[\widehat{x}_i(h) - x_i(h)\right]^2 \right\}^{1/2}. \tag{13}$$

Here $N = 1{,}296$ for synthetic confirmation and $N = 200$ for NGSIM, and H counts frames at $\Delta t = 0.1$ s. Thus "1-s RMSE" is the root mean square over frames $1, \dots, 10$, *not* just the position at $t = 1$ s. The 0.5, 1, and 2-s windows overlap. The recorded NGSIM position remains an observed proxy, not latent ground truth.

2.7. Two complementary experimental cases

Known-truth simulation. In the completed Experiment005, we generated 1,296 confirmatory analytic trajectories spanning six jerk regimes, three observation-noise levels and three error structures, with 80 historical observations and 20 clean future observations at 10 Hz. Historical jerk is scored against the *same discrete operator* applied to the true clean positions, not against an unmatched instantaneous derivative. Experiment006 compared fixed and estimated references and Gaussian parameter selection on these preserved cases. The new Experiment008 compares the PLS and TSVD filters with splines and Savitzky–Golay on *exactly these archived simulations*, without regenerating them or refitting the original PLS parameter search. Experiment007 separately generated 1,944 paired noise–dynamics calibration conditions, based on 72 independent innovation parents; its known-covariance analysis is described in Section 4.

Real recorded trajectories. The Federal Highway Administration/USDOT NGSIM dataset (DOI: 10.21949/1504477) supplies camera-derived longitudinal vehicle position at 10 Hz (U.S. Department of Transportation, Federal Highway Administration, 2016). Source `Local_Y` values are converted from feet to meters by 0.3048. The initial descriptive illustration used 15 overlapping chronological windows from five US-101 track segments that contain only two distinct integer vehicle IDs. A larger fixed-source follow-up selected 200 quality-qualified episodes (100 US-101 and 100 I-80); their IDs cannot be equated with 200 unique physical vehicles. Every episode supplies 80 historical positions and withholds $h = 5, 10, 20$ future frames at a prespecified origin. The new TSVD comparison reuses the *same* archived 200 episodes and training-only PLS parameter choices. Real jerk truth and sensor precision are *not independently observed*. The 200 episodes represent five coarse recording-time blocks, so pooled forecasting scores describe this convenience sample, not independent vehicle-population inference. Source queries, file checksums, configuration and raw/fitted outcome rows remain documented separately for every experiment.

3. Simulation case: reconstruction of known physical jerk

3.1. What was simulated and what was held fixed

The known-truth simulation contains 1,296 frozen confirmatory trajectories from the six prespecified vehicle jerk regimes: zero jerk, positive and negative constant jerk, sinusoidal jerk, abrupt jerk reversal and a finite braking-related jerk pulse. At each of three position-noise levels (0.03, 0.12, 0.30 m) we consider independent observation errors, serially correlated errors, and terminal contamination. Every trial supplies 80 noisy historical positions, the original clean position and its known discrete third-difference jerk, and 20 clean future positions. The pilot and confirmation use distinct random seeds; the confirmation trajectories are precisely the original Experiment005 data, *not* regenerated for the comparisons below.

Experiment006 examined the choice of the reference mean and smoothing penalty on this data. Its global cubic OLS reference with continuous profile Gaussian likelihood attained 2.441 m/s^3 historical discrete-jerk RMSE; the corresponding fixed-zero reference attained 2.587 m/s^3, while joint unknown-mean REML attained 3.295 m/s^3. The latter result is not paradoxical: maximum restricted likelihood is not an oracle minimizer of physical-jerk reconstruction loss.

3.2. Soft PLS versus hard SVD truncation and conventional smoothers

We now compare PLS against the *different estimators* in Eqs. (7)–(8), alongside GCV cubic smoothing splines and a 21-point degree-three Savitzky–Golay filter. For PLS, the $\widehat{\mu}$ and continuous profile-likelihood λ are read *unchanged* from the successful Experiment006 archive. The truncated-SVD methods use this same training-only cubic reference mean: one truncation rank is selected from training data using GCV, and the other is matched to the PLS effective degrees of freedom. The matched-rank method is an *ablation* of soft versus hard spectral shrinkage, not a separate statistically optimal rank estimate.

The matched-complexity comparison isolates an advantage of soft PLS shrinkage over a hard roughness-spectral cutoff in the tested synthetic jerk-reconstruction task. By contrast, SVD rank chosen to fit observed positions through GCV is extremely unstable for *third-difference jerk*: its historical jerk RMSE is 250.718 m/s^3, although its first-second clean-position window RMSE (0.489 m) is not comparably disastrous. This distinction is evidence that accurate position reconstruction and accurate *higher-order differentiation* are different objectives. The archived per-trajectory rows identify a concentrated failure: TSVD-GCV selected rank 75 or 76 of the 77 positive singular directions in 145 of 1,296 trajectories (11.2%). These cases contributed 78.9% of that estimator's aggregate squared historical jerk error. This is an accounting of the *existing* Experiment008 outcomes, not a new independent experiment or a separate selection of particularly favorable trajectories. Moreover, its rank criterion conditions on the training-estimated cubic reference $\widehat{\mu}$, as explained in Section 2; the label "TSVD-GCV" denotes that specific conditional plug-in score, not exact GCV for the fully data-dependent procedure.

The GCV cubic spline and Savitzky–Golay comparators likewise smooth positions, but their respective penalties and local window mechanisms do not directly impose the PLS reference constraint $D_3\tau \approx \mu\mathbf{1}$. The experiment does *not* establish that every spline, window, or SVD denoising procedure is inferior to PLS; only these prespecified training-based configurations were evaluated.

We therefore distinguish the established *smoothness-index interpretation* from the empirical quality of estimated derivatives, and neither from the separate extrapolation rule. The calibration experiment in Section 4 is deliberately reproduced on a different factorial design with separate known observation-noise and jerk-innovation interventions; the two simulation studies must not be pooled as if they shared one sampling model.

4. Smoothness as relative noise attribution: calibrated simulation evidence

4.1. What the normalized index identifies

For a third-order difference operator $D = D_3$ and a fixed reference μ, the model $y = \tau + \varepsilon$, $D\tau = \mu\mathbf{1} + u$, with independent Gaussian disturbances $\varepsilon \sim N(0, \sigma_\varepsilon^2 I)$ and $u \sim N(0, \sigma_u^2 I)$, implies

$$\lambda = \frac{\sigma_\varepsilon^2}{\sigma_u^2}, \quad S(\lambda) = \frac{1}{m}\sum_{j=1}^{m} \frac{\lambda\delta_j}{1+\lambda\delta_j}, \quad m = n-3. \tag{14}$$

On the spectral coordinates of Dy, the jth summand is the *fraction of variance due to differenced observation error*, not the fraction of observations that are erroneous. Figure

Table 1
Same 1,296 frozen simulated trajectories: accuracy against known discrete jerk in the 80-point historical window, and future *clean* position RMSE averaged over the first ten future frames (through one second). The future-position comparison uses the same last-25-point quadratic continuation for all fitted trends. True jerk is scored with the same third-difference operator $D_3/(\Delta t)^3$ applied to the clean positions. Smaller values are preferable; the columns describe *different physical targets*.

Historical trend estimator	Jerk RMSE (m/s^3)	First-1-s position RMSE (m)
PLS, cubic OLS $\hat{\mu}$ + profile ML	**2.4413**	**0.4353**
SVD truncation, matched effective complexity	2.9416	0.4502
SVD truncation, training-only GCV rank	250.7181	0.4889
Natural cubic smoothing spline, training GCV	60.9131	0.4614
Savitzky–Golay, 21-point cubic fit	36.8291	0.4825

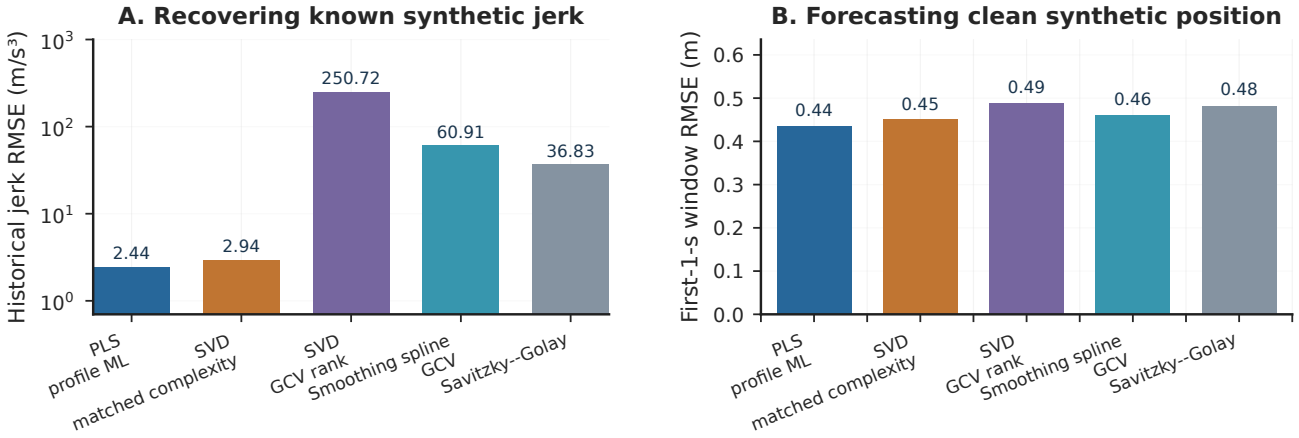


Figure 1: Known-truth reconstruction and clean-future forecasting from the same archived 1,296 simulated trajectories. (A) Historical third-difference jerk RMSE on a logarithmic vertical scale (m/s^3); (B) clean-position RMSE across the first ten future frames under the shared 25-point continuation (m). Truncated SVD is a hard spectral-filtering *estimator*, not a different implementation of the PLS linear solve. The GCV-selected truncation can retain high-frequency components that strongly amplify differentiation errors. Different plot axes and units must not be interpreted as a single aggregate score.

3 illustrates these spectral weights and the exact conditional complexity identity $\mathrm{df}(H_\lambda) = n - (n-3)S(\lambda)$. This latter identity holds for a *fixed* reference mean; if μ is estimated, the unpenalized effective complexity can increase.

If observation noise and genuine jerk innovations are correlated in time, the scalar variance-ratio formula no longer gives the correct covariance allocation. Nevertheless, the known generative covariance matrices define a target for validation:

$$S_{\mathrm{cov}} = \frac{1}{n-3}\,\mathrm{tr}\left[(C_u + C_\varepsilon)^{-1} C_\varepsilon\right], \tag{15}$$

where $C_u = \mathrm{Cov}(D\tau - \mu\mathbf{1})$ and $C_\varepsilon = \mathrm{Cov}(D\varepsilon)$. Under independent Gaussian disturbances Eq. (15) equals Eq. (14) at the true variance ratio. Under covariance misspecification an estimated S need not estimate S_{cov}.

4.2. A known-jerk experiment with paired noise interventions

We independently manipulate position-measurement noise and the actual fluctuations of physical jerk. Each of $n = 80$ clean positions is constructed from a prescribed true third difference $D_3\tau = (\Delta t)^3 j$. Experimental levels are three observation-noise standard deviations $(0.01, 0.05, 0.20\,\mathrm{m})$ and three jerk-innovation standard deviations $(0.5, 2, 8\,\mathrm{m/s^3})$. We cross these with independent, AR(1), and terminal-contaminated measurement errors, and with independent, AR(1), and abrupt-reversal jerk processes. The terminal outlier has randomized sign, so its mixture covariance is known, but the observations are not Gaussian. The reversal violates the constant jerk-reference assumption.

The frozen confirmatory design has 1,944 factorial cases and four smoothing selectors per case. *Only 72 independent parent random draws underlie the 1,944 cases*: standardized jerk innovations and measurement-error draws are held constant across all 27 treatments within each parent. The 162-case pilot uses a separate seed namespace. All 7,776 confirmatory fits were valid. The four selectors are Gaussian profile likelihood with the known generative μ (a simulation-only control), unknown-mean Gaussian REML, conditional profile likelihood with noisy global cubic OLS $\hat{\mu}$, and zero-mean GCV on the original 15-point smoothness grid.

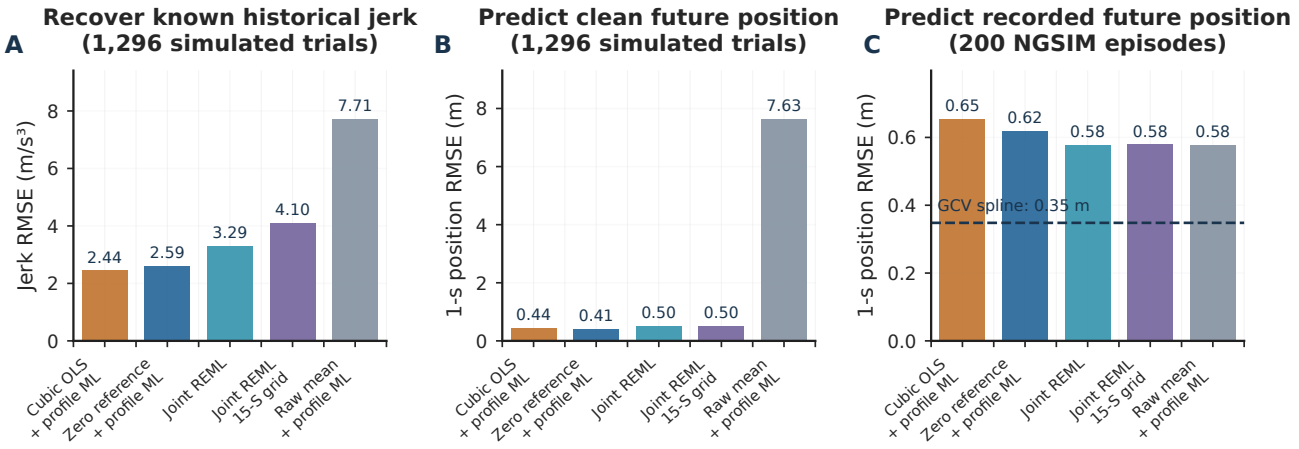


Figure 2: Additional preexisting Experiment006 reference-mean and parameter-selection comparison. (A) Historical known-discrete-jerk RMSE in m/s^3; (B) first-second clean-future position-window RMSE in the same synthetic trajectories; and (C) separately recorded future-position RMSE from 200 real NGSIM episodes, with the spline benchmark shown for context. The different targets, data generating processes and selected forecast rules explain why the apparent best method depends on the question. No real jerk ground truth exists in (C).

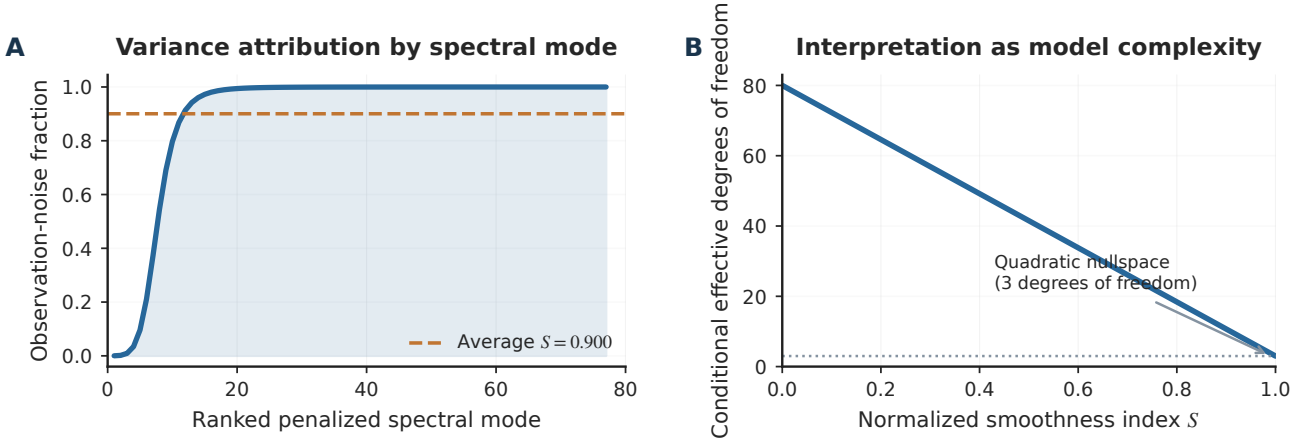


Figure 3: Meaning of the normalized $d = 3$ smoothness index. (A) Ranked observation-noise variance fractions in 77 penalized third-difference spectral directions for $n = 80$, $\Delta t = 0.1$ s, observation SD 0.05 m and true jerk innovation SD 2 m/s^3 under the independent Gaussian model; their average is S. (B) Exact conditional effective degrees of freedom as a function of S when the mean of differences is fixed. Neither panel treats S as an absolute sensor-noise standard deviation.

4.3. Calibration and independent changes in noise and jerk

Under the correctly specified independent Gaussian model, unknown-mean REML produced Spearman correlation 0.965 between selected S and the true spectral noise fraction, with absolute calibration error 0.0115 averaged over the 216 paired factor combinations (24 stochastic parents). The known-reference control achieved 0.967 and 0.0107, respectively, while the global cubic OLS plug-in achieved 0.966 and 0.0143. The zero-reference GCV comparator achieved 0.901 and 0.0249. These are *descriptive* correlations over dependent factorial cases, not independent sample significance tests.

For unknown-mean REML under independent Gaussian errors, raising position-noise SD from 0.01 to 0.20 m at fixed jerk SD increased fitted S by 0.1498 (known theoretical increase 0.1451). Raising actual jerk innovation SD from 0.5 to 8 m/s^3 at fixed observation noise *decreased* fitted S by 0.1387 (known theoretical decrease 0.1337). Paired parent-level bootstrap intervals were $[0.1454, 0.1544]$ and $[-0.1428, -0.1345]$, respectively, conditional on this particular simulation design. Figure 6 shows the corresponding results across all nine covariance and maneuver regimes, including the less accurate misspecified cases.

A direct identifiability counterexample prevents interpreting S as *absolute measurement precision*: increasing $(\sigma_\varepsilon, \sigma_j)$ from (0.05 m, 2 m/s^3) to (0.20 m, 8 m/s^3) preserves the variance ratio and therefore S under the independent Gaussian model, even though the observation-noise variance increases sixteenfold. The same-ratio experiment produced

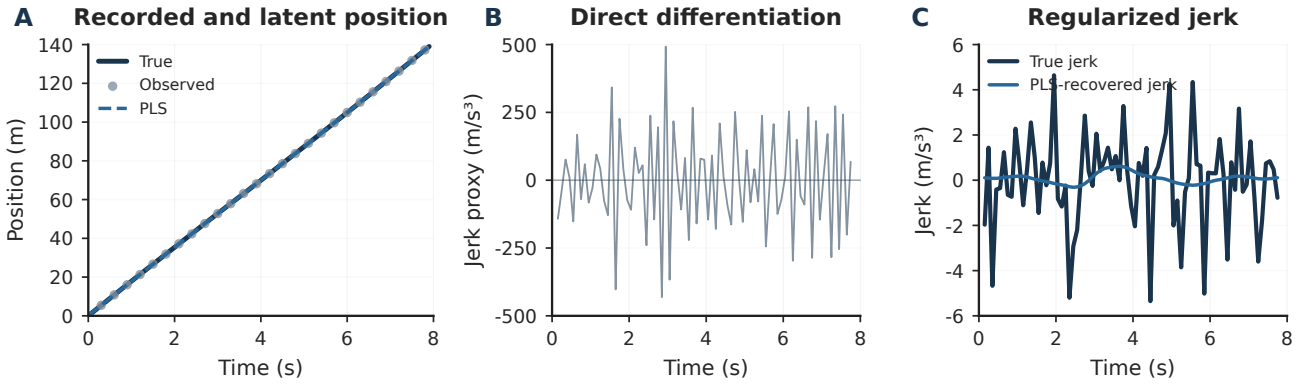


Figure 4: One *predeclared* analytic trajectory from the independent Gaussian regime (confirmation trial 000010; observation SD 0.05 m, jerk-innovation SD 2 m/s^3). (A) Prescribed clean position, noisy observations and PLS fit using global cubic OLS $\hat{\mu}$ and profiled Gaussian likelihood; (B) the raw third difference divided by $(0.1\,\mathrm{s})^3$, with severe noise amplification; (C) known prescribed jerk and the PLS-reconstructed third difference. The raw and regularized jerk panels have different vertical scales. This generated example is *not* a claimed independent real NGSIM jerk measurement.

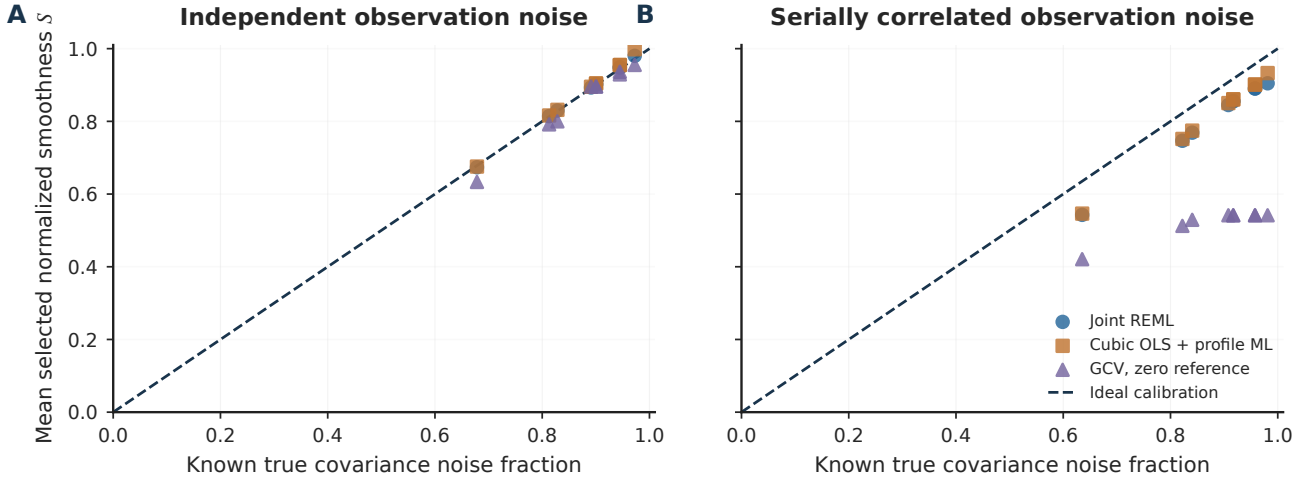


Figure 5: Empirical calibration of the selected smoothness coordinate against the *true covariance-based* noise fraction. Each point aggregates 24 independent parents at one of nine observation-noise/jerk-volatility combinations; three practical selectors are shown. (A) Independent observation noise and independent jerk innovations. (B) Correlated AR(1) observation noise with otherwise independent jerk innovations: rank association persists, but calibrated values systematically depart from the identity line. An oracle known-μ control is excluded from this practical-selection visualization.

an estimated mean ΔS statistically indistinguishable from zero by design. The index is relative to dynamical innovation variation, not a stand-alone metrological rating of recording quality.

4.4. Where the precision interpretation fails

Figure 7 displays calibration errors under deliberately misspecified noise and dynamics. For joint REML, mean absolute error increased from 0.0115 under independent Gaussian disturbances to 0.0712 when the measurement error had AR(1) correlation of 0.65, and to 0.1111 when both the measurement noise and jerk innovations were serially correlated. Hence favorable rank correlation alone is insufficient to claim quantitatively accurate noise attribution.

This independent calibration experiment characterizes the conditional meaning of selected S, not which smoother best recovers physical jerk or forecasts position. The preceding known-jerk experiment and the subsequent recorded-vehicle case examine those separate targets; their observations and sampling units must not be pooled as independent replications.

5. Real-data case: recorded vehicle trajectories

5.1. Observed data are not known true jerk

The Federal Highway Administration's NGSIM US-101 and I-80 trajectories supply camera-derived longitudinal position, not independent noise-free physical position or jerk.

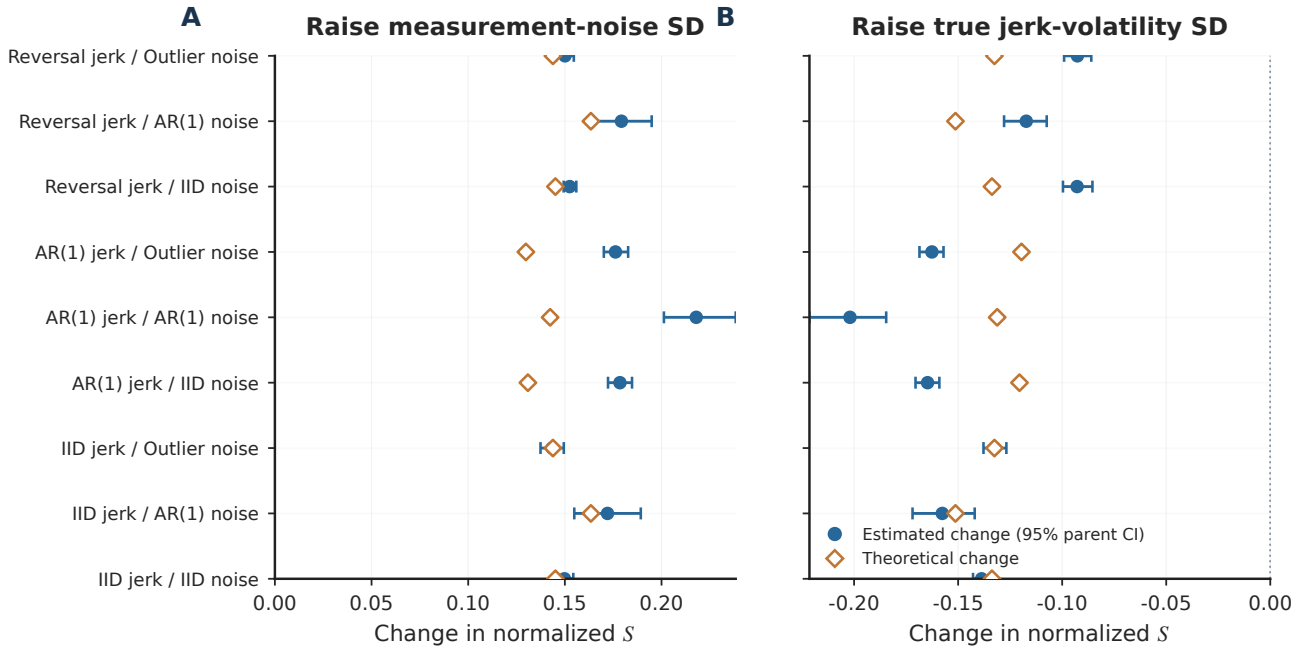


Figure 6: Two separately controlled interventions for selected joint-REML smoothness across nine jerk/measurement-error structures. (A) Increasing observation-noise SD with true jerk volatility fixed; (B) increasing true jerk volatility with observation noise fixed. Blue circles and whiskers show parent-cluster means and 95% Monte Carlo bootstrap intervals; open orange diamonds are known covariance-theoretic effects. Theoretical effects vary by correlation regime. Paired factor combinations within each independent parent are never resampled as independent observations.

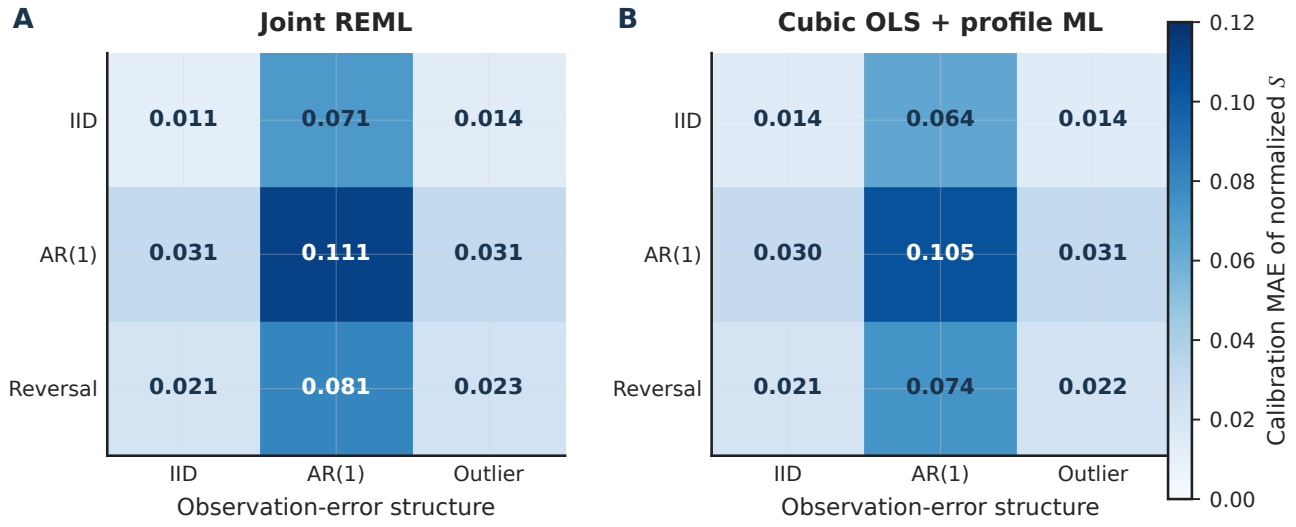


Figure 7: Absolute S calibration error against the *known* covariance-noise fraction, by true jerk structure (rows) and observation-error structure (columns). Left: properly mean-adjusted joint REML; right: global cubic OLS reference followed by fixed-mean Gaussian profile likelihood. The independently Gaussian regime occupies the upper-left cell. Darker values indicate larger errors; the AR(1) observation-error cases are conspicuous failures of the independent-noise statistical model.

For an initial visualization we retain the five prespecified US-101 track segments from the original case study, comprising 15 overlapping training/forecast origins and only two distinct integer vehicle IDs. The more informative recorded-position validation uses 200 preselected NGSIM episodes (100 from each site), each with 80 past 0.1-s observations and 5, 10 and 20 future frames withheld. The source and selection rules were frozen before model comparisons. These episodes correspond to 113 distinct (site, integer vehicle ID) pairs and five coarse recording-time blocks; repetitions of an integer identifier need not be the same physical driver. Our

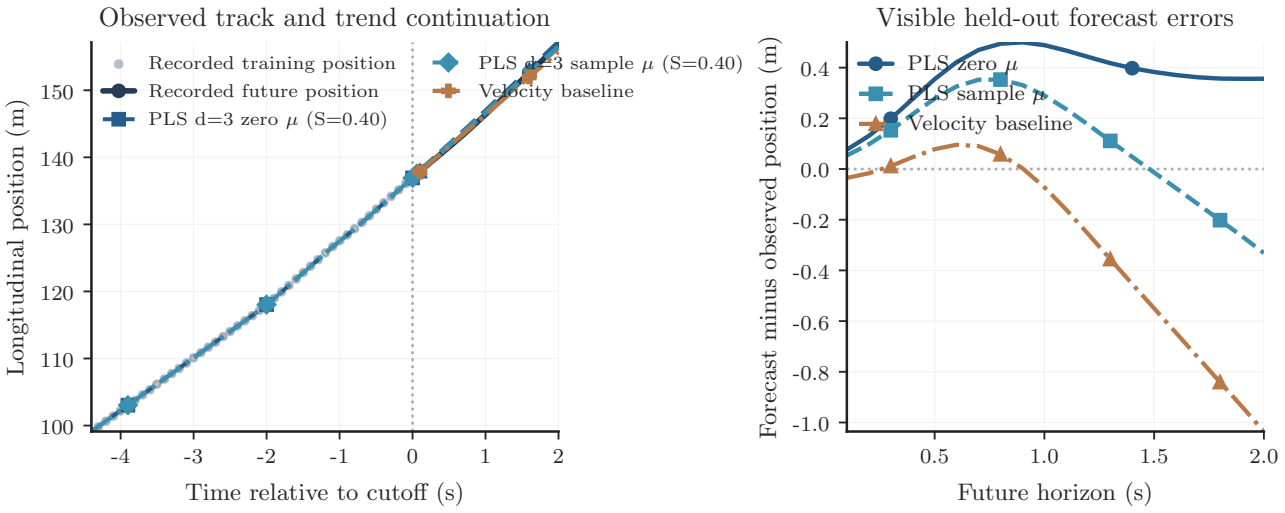


Figure 8: Predeclared chronological US-101 illustration: the measured historical segment, withheld future recorded positions, fitted third-order PLS trends with zero and training-derived reference means, and their historical-trend extrapolations. No future observation enters historical fitting or smoothing-parameter selection. This pilot figure is not drawn from the larger 200-episode sample to claim an additional independent test.

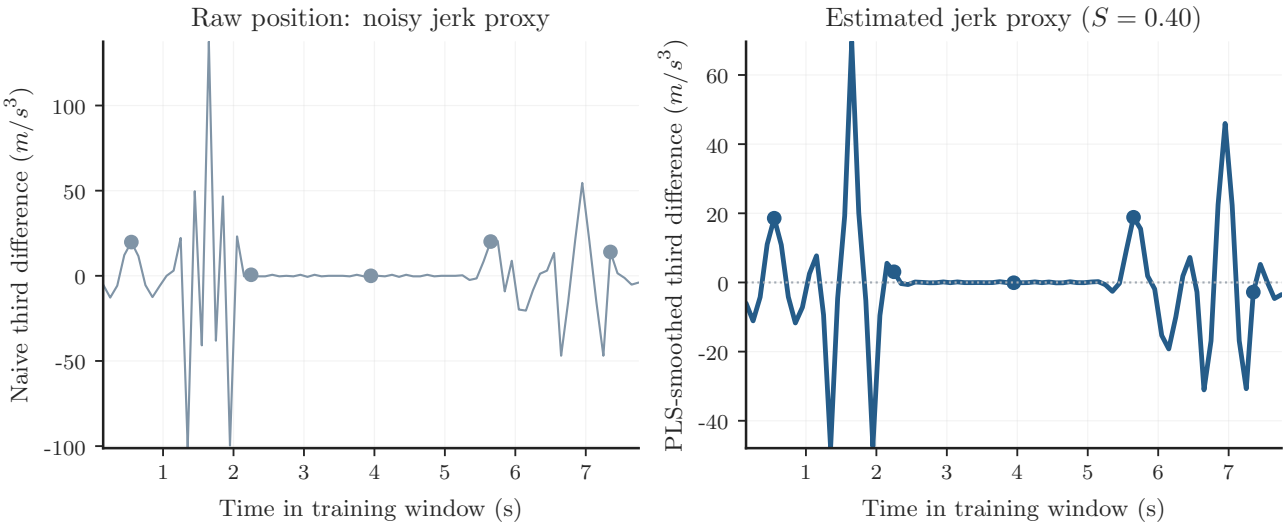


Figure 9: Raw and PLS-regularized *proxies* for third differences of the same measured US-101 position segment, scaled by $(0.1\,\mathrm{s})^3$ to have jerk units. The panels use different axes. A smoother curve is not proof of physical jerk accuracy: this real recorded trajectory has no independently measured true jerk.

pooled error summaries are descriptive and do not justify treating 200 episodes as independent population draws.

5.2. Matched PLS versus spectral SVD and physical baselines

The SVD comparison is based on the same original 200 recorded episodes and reuses archived, training-selected global cubic OLS references and continuous Gaussian profile-ML penalties. SVD-GCV chooses its rank using historical positions only under the conditional plug-in criterion in Eq. (9). The matched-rank SVD equates *nominal fixed-reference* effective complexity with PLS; it does not exactly match the full data-dependent estimator's degrees of freedom. The same three-point continuation *algorithm* in Eq. (12) is applied to each historical fit. Its reference μ is the common training-derived cubic-OLS value for PLS and TSVD, but zero for spline and Savitzky–Golay fits. Thus the boundary rule is shared, not every model assumption or third-difference reference. The comparison is not a benchmark of Cholesky versus SVD run time.

In the new matched-complexity comparison, PLS and hard-truncated SVD produce very similar pooled first-second recorded-position window RMSE (0.5144 and 0.5118 m). This real-data result does *not* establish superiority of PLS over SVD on forecasting, despite its stronger known-jerk

Table 2
Future *recorded-position* RMSE (m) averaged over frames 1 through $H = 5, 10, 20$ in each of 200 selected NGSIM episodes, with chronological 80-point training windows. Values are *cumulative-window* errors (Eq. (13)), not terminal-frame errors. The first five rows use the common three-point constant-third- difference continuation; the final GCV spline row uses its *own* quadratic Taylor continuation and is therefore an end-to-end method comparison, not an identical boundary-rule ablation. No independent true jerk is available.

Historical smoother + continuation	$H = 5$	$H = 10$	$H = 20$
PLS, cubic OLS reference + profile ML	0.1434	0.5144	1.8829
SVD cutoff, matched complexity	0.1423	0.5118	1.8787
SVD cutoff, training-only GCV rank	0.1512	0.5337	1.9339
Natural cubic smoothing spline, GCV	0.1393	0.4946	1.7965
Savitzky–Golay (21,3)	0.3464	0.8188	2.2822
GCV natural cubic spline + quadratic Taylor	**0.1354**	**0.3480**	**0.8787**

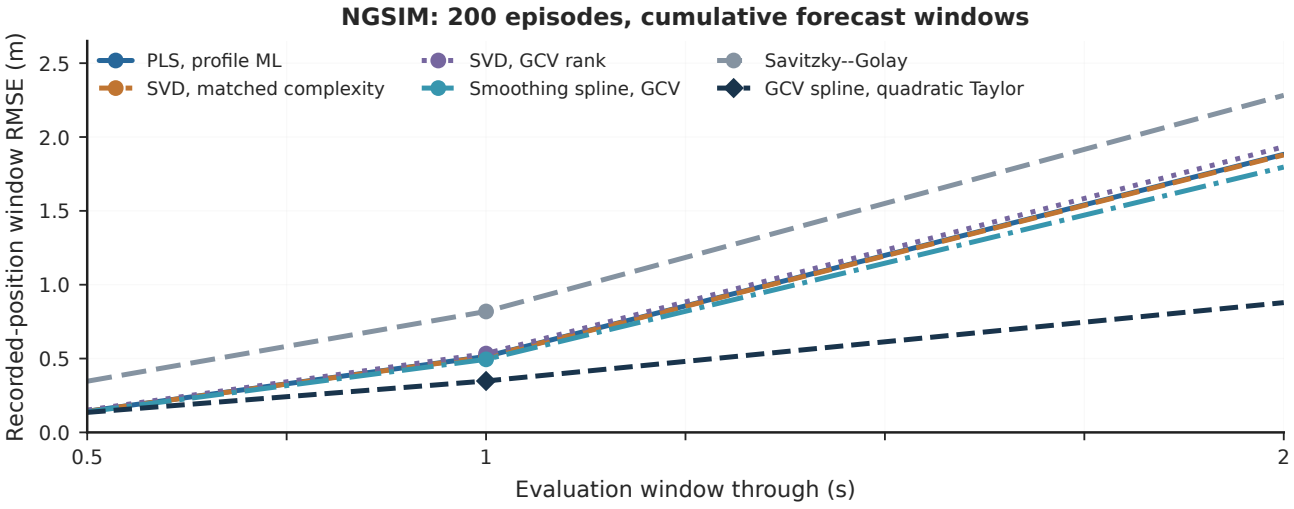


Figure 10: Recorded-future longitudinal position cumulative-window RMSE across 200 real NGSIM episodes and three evaluation windows (0.5, 1 and 2 s). Five historical smoothers share the same last-three-point continuation, with the independently selected GCV cubic spline and quadratic Taylor forecast shown as an additional dashed-line *end-to-end* benchmark. The recorded future is the only observable target; no latent position or true jerk appears in these errors.

reconstruction performance in Section 3. The GCV cubic smoothing spline with quadratic Taylor extrapolation achieves smaller pooled recorded-position window RMSE at each horizon. The comparison changes both the historical smoother and its continuation; it cannot attribute the entire difference to spline smoothing alone. An alternative continuation of the *same fitted historical trend* can therefore matter as much as the smoother.

5.3. Original historical illustration and forecast horizons

The smaller original exploratory study compared PLS under several difference orders, regularization criteria and μ choices, then examined a last-three-point $d = 3$ continuation. Those original numerical tables and all four-order criterion heatmaps are retained in the research archive rather than treated as the main evidence of physical superiority.

In the original 15-origin comparison, PLS with zero reference and the three-point $d = 3$ recurrence attained 0.0745 m pooled observed-position RMSE at 0.5 s, whereas the GCV spline's quadratic continuation attained 0.1233 m. At 2 s, the spline attained 0.8085 m versus 1.2998 m for that PLS recurrence. These rankings are *exploratory*; the larger 200-episode study demonstrates how small-sample method comparisons can change with different recorded cases.

6. Discussion and conclusions

The main question of this work is how to extract a physically interpretable third-difference trend from measured position, not how quickly competing matrix factorizations solve the *same* penalized least-squares equations. The established PLS estimator provides a direct penalty on deviations of discrete jerk from a reference mean and an intuitive normalized smoothing coordinate S.

Under the independent Gaussian variance-ratio model, S equals the mean spectral fraction of observed third-difference covariance attributable to observation error. The controlled paired calibration experiment supports this interpretation quantitatively when the covariance assumptions

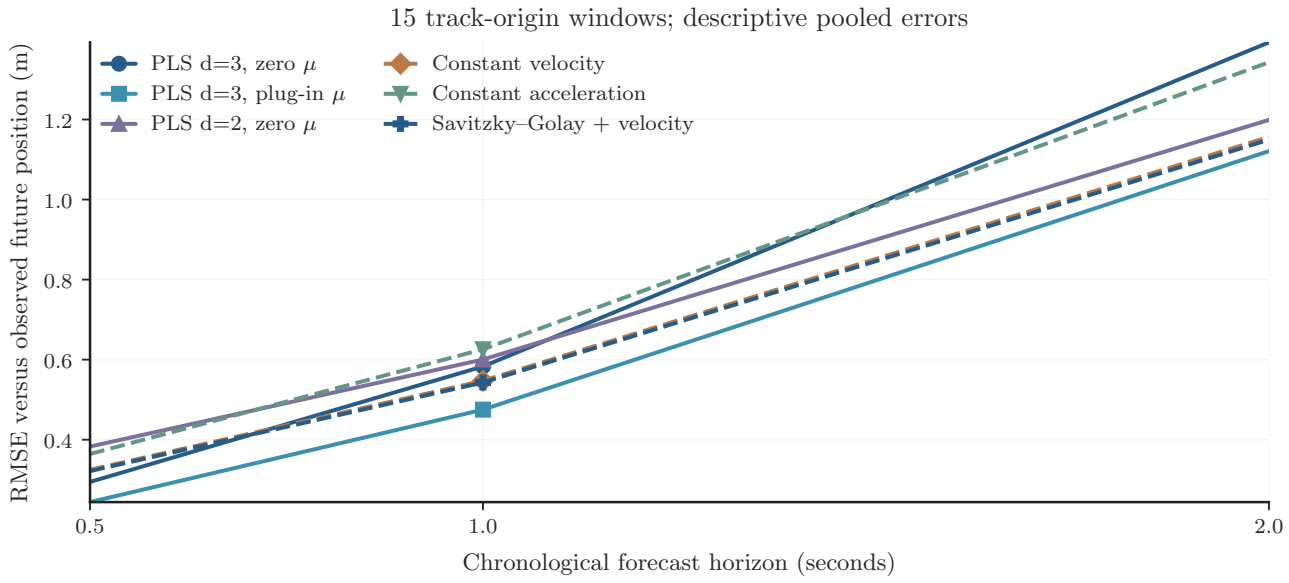


Figure 11: Original exploratory chronological comparison over 15 overlapping forecast origins from five US-101 track segments, including PLS and conventional constant-velocity, constant-acceleration and Savitzky–Golay baselines. The plotted horizons use measured future positions, not known underlying truth. This smaller sample is not independent of the historical case-study provenance, nor representative of the larger 200-episode follow-up.

hold. Its stress tests also identify a clear failure: serially correlated measurement noise can bias S substantially even when its rank association with relative noise persists. No value of S alone can identify absolute observation-noise precision, since scaling both genuine jerk volatility and measurement error by the same amount preserves their variance ratio.

A second known-truth simulation case provides a different outcome: PLS with the historical cubic-OLS reference and continuous Gaussian profile likelihood has lower discrete-jerk RMSE than a hard spectral SVD cutoff at nominally matched fixed-reference degrees of freedom, and much lower RMSE than the position-GCV truncation in the tested regimes. Hard cutoff and continuous shrinkage are *different estimators*; using a full SVD in place of a Cholesky factorization of the PLS normal equations would not give such a statistical comparison. The surprising sensitivity of GCV-truncated SVD to third differentiation illustrates that selecting a smooth position path and recovering higher-order derivatives need not be equivalent goals. This result is conditional on the tested spectral basis, reference mean policies and noise-generating processes; it is not a universal rejection of SVD-based denoising, smoothing splines, or local filters.

The real recorded-position results are more cautious. Across 200 quality-qualified NGSIM episodes, the three-point PLS and effective-complexity-matched SVD forecasts are similar; a GCV natural cubic smoothing spline with quadratic Taylor continuation has the lowest pooled recorded-future position window RMSE among the reported procedures. The smaller original US-101 exploratory sample favored the PLS recurrence at 0.5 s; its five segments came from only two integer vehicle identifiers and cannot establish a general forecasting advantage. The larger sample contains multiple shared recording-time blocks, not 200 demonstrably independent physical drivers. True jerk and latent clean position are not measured in NGSIM. Consequently forecast accuracy against camera positions is not evidence of physical jerk accuracy.

The reference third-difference mean μ also matters: a nonzero global cubic OLS reference can improve simulated jerk reconstruction without making joint Gaussian REML or a particular future extrapolator universally optimal. The last-three-point recurrence assumes constant third difference *after the boundary*, and does not provide a predictive distribution or anticipate unobserved driver actions. These differences explain why smoothness calibration, jerk recovery and future-position forecasting should be assessed as *separate* estimands.

Our contribution is therefore a reproducible physical application and conditional characterization of third-order PLS and its interpretable smoothness index. We do not claim to have invented PLS, third-order penalties, the smoothness index, or SVD truncation. Future work should use additional independent recording scenes, credible external jerk/position ground truth, experimentally characterized sensor-error covariance, and competing trajectory/state-space methods. Numerical backend timing, optimized factorization strategies and many-parameter selection tensors belong in separate computational-statistics manuscripts rather than this physical case study.

Reproducibility and data rights

The US DOT/FHWA NGSIM dataset (DOI: 10.21949/1504477) is credited as the sole source of observed positions; the dataset portal identifies a CC BY-SA 3.0 license. Source query and extraction rules, tracked vehicle identifiers, segment starting timestamps, immutable checksums, chronological forecast rows, solver diagnostics, and original TeX-backed PGF figure sources are preserved under the case-study directory. The new Experiment006/007 synthetic and NGSIM follow-up results are preserved separately, including complete generated ground truth, observation errors, selected smoothness values, paired inference, reproducible vector plots, Conda environment records, exact file paths and SHA-256 manifests. The original official-position subset has not been artificially contaminated or pre-smoothed. Figure-derived jerk is explicitly estimated rather than observed.

Declarations

This is an illustrative research draft. Authorship, correspondence, funding, conflicts and any journal submission require coauthor review and agreement.

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